

a.Introduction to WAN Technologies : Definition of WAN, Types of WAN Connections, WAN Devices: CSU/DSU, WAN Aggregators.

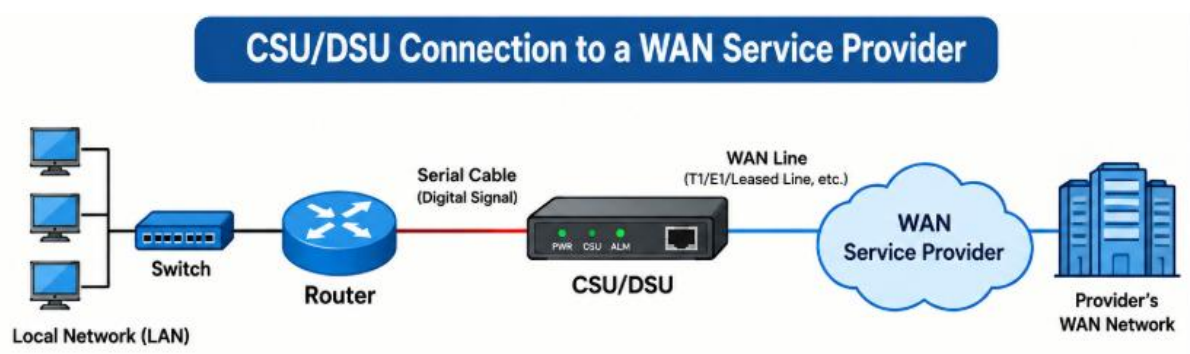
Task 1.

Create a table comparing at least three types of WAN connections (such as leased line, DSL, and MPLS) with columns for speed, cost, and typical use cases.

WAN Connection	Speed	Cost	Typical Use Case
Leased Line	High and dedicated; commonly from Mbps to Gbps	High	Banks, large offices, data centers, organizations needing reliable dedicated connectivity
DSL	Usually Mbps to hundreds of Mbps, depending on technology/provider	Low to Medium	Small offices, homes, small businesses, Internet access
MPLS	Typically Mbps to multiple Gbps	High	Enterprises connecting multiple branch offices securely and reliably
Internet / Broadband	Mbps to Gbps	Low to Medium	Branch offices, cloud access, general Internet connectivity

Task 2.

Draw a labeled diagram (on paper or using any drawing tool) showing how a CSU/DSU connects a router to a WAN service provider, and explain the role of each device in 2-3 sentences.



Router:

The router connects the local network to the WAN and determines where packets should be sent. It forwards traffic between the customer's LAN and the WAN service.

CSU/DSU:

The CSU/DSU connects the router's digital interface to the service provider's WAN circuit. It performs functions such as signal conversion, timing/clocking, and line termination.

WAN Service Provider:

The provider supplies the WAN circuit that carries data between geographically separated locations.

Task 3.

List three real-world examples of popular apps or services (like IRCTC, WhatsApp, or Zomato) that rely on WAN connections, and briefly describe how WAN enables their functionality.

1. IRCTC

IRCTC uses network connectivity between its servers, railway systems, payment services and users across different locations. WAN connectivity allows users in different cities to access railway reservation services.

2. WhatsApp

WhatsApp relies on wide-area Internet connectivity to exchange messages, voice calls, video calls and media between users in different cities and countries.

3. Zomato

Zomato uses WAN/Internet connectivity to connect customers, restaurants, delivery partners and its cloud/server infrastructure. This allows orders, locations, payments and delivery updates to be exchanged in real time.

Task 4.

Research online and find an image or datasheet of a physical WAN aggregator device used in enterprise networks. Paste the image or link, and write two key features of that device.

Official Cisco datasheet:

Cisco ASR 1000 Series Aggregation Services Routers Data Sheet

Cisco describes the ASR 1000 Series as an aggregation platform that can aggregate multiple WAN connections and network services, with line-rate forwarding from 2.5 to 200 Gbps.

Two key features

1. WAN aggregation:

It can combine/aggregate multiple WAN connections and network services, making it suitable for enterprise and service-provider networks.

2. High performance:

The ASR 1000 Series supports high-speed forwarding from 2.5 Gbps up to 200 Gbps, making it suitable for large enterprise WAN and data-center environments.

